Audio Codec Effects on Feature Attribution in Speech Emotion Recognition

Akshath R

***Abstract*—Speech emotion recognition (SER) systems are usually validated on uncompressed audio but deployed on coded audio. This study examines how coding affects the evidence a model uses, beyond its accuracy. Fifteen classifiers were fitted to 436 named acoustic descriptors from CREMA-D, with every clip rendered as WAV, as MP3 and AAC at 64 kbit/s, and as an AAC re-wrap that changes only the container. Cross-format losses arise from train/serve mismatch rather than lost information: an RBF-SVM trained on WAV reaches 0.203 balanced accuracy on MP3, but 0.589 when trained on MP3, against 0.586 on clean audio. Under mismatch the neural network stops predicting three of the six classes, a failure that balanced accuracy conceals. Entropy trees fitted per condition keep the same root descriptor but reproduce the reference structure only to depth 2 under MP3 and depth 0 under AAC, compared with depth 3 for the empty container control, while extracting the same total information gain to within 2 %. The empty re-wrap alone displaces 8.4 of a tree's 25 most important descriptors; against this floor, the codec-specific displacement is significant ($p < 0.002$) but 2.6 times smaller than the uncalibrated figure. Attribution-shift claims should therefore be reported against a matched empty-perturbation control.**

***Index Terms*—Speech emotion recognition, explainable artificial intelligence, perceptual audio coding, feature attribution, decision trees, information gain, covariate shift, null calibration.**

I. Introduction

AUDIO rarely reaches an inference endpoint in the format in which it was captured. Contact-centre archives store MP3, streaming clients negotiate Opus or AAC, and broadcast pipelines resample and requantise audio. Benchmarks, in contrast, are distributed as uncompressed WAV and read directly from disk. The difference is usually considered harmless because every ingestion path decodes to linear PCM before features are extracted. Where this assumption has been tested, it has been tested on accuracy, with mixed results: Reddy and Vijayarajan [1] report codec effects on speech emotion recognition (SER) that are real, modest and dependent on the front end.

Accuracy alone is an incomplete measure for a system that must report both *what* it concluded and *why*. A deployed affective system may present an attribution over acoustic evidence to a clinician, an auditor or a quality-assurance analyst, and that second output has robustness properties of its own. Ghorbani et al. [2] showed for image classifiers that inputs which are indistinguishable to a human observer and receive identical predictions can still receive very different attributions, and Alvarez-Melis and Jaakkola [3] reported the same fragility in a more general setting. Both studies used adversarial perturbations. Transcoding is not adversarial, but it is routine, widespread and seldom logged.

Studying the effect of coding on evidence requires a model whose evidence can be inspected. An attribution placed on a latent dimension of an end-to-end audio model does not correspond to a quantity an acoustician would recognise. A model fitted to *named* descriptors in the GeMAPS [4] and openSMILE [5] tradition does: an attribution on `prosody_cpps` is a statement about breathiness, and a split on `spectral_contrast_6_mean` is a statement about the 6 kHz to 8 kHz band. The study is restricted to this setting so that every structural result is exact rather than estimated and can be checked against an acoustic interpretation.

A claim that compression changed which features a model relies on is easy to make and difficult to justify. Feature-importance rankings are unstable: two descriptors that split the data almost equally well can exchange places, and a single change near the root of a tree alters every node below it. Adebayo et al. [6] and Krishna et al. [7] have raised general concerns about attribution reliability, but measuring the shift caused by a manipulation that carries no information is not yet standard practice. The present design includes such a manipulation, namely the same codec output routed through two different containers. Once that floor is measured, most of the apparent codec effect on importance rankings is attributable to generic sensitivity to perturbation.

The contributions of this paper are as follows.

1) A matched-format decomposition of the codec effect. Training and serving in the same coded condition recovers a cross-format loss of up to 38.3 balanced-accuracy points, which shows that $64\,\mathrm{kbit\,s^{-1}}$ coding moves the affective information outside the fitted range without removing it. A per-class analysis shows that the mismatched network abandons half of the label space, which balanced accuracy cannot reveal (Sections V-B–V-D).
2) Per-condition tree structure as an exact attribution. Entropy trees fitted independently per condition identify which test the codec changes, at the level of a named descriptor and a threshold. A depth profile quantifies how far that agreement extends, and evaluating information gain at every node separates structural from informational agreement (Sections V-E–V-F).
3) Null calibration for shifts in feature importance. Three floors (tie-breaking, actor resampling, and an informationally empty container re-wrap) are measured, and a paired within-draw design separates the codec effect from all three (Section V-G).

A. R is an independent researcher (e-mail: akshath.r333@gmail.com).

The complete feature inventory and the per-condition tree splits are given in the supplementary material.

4) A repair that can be diagnosed and costs little. Neutralising a single named descriptor recovers 98 % of a kernel machine's loss and 92 % of a neural network's loss with no cost on clean audio (Section V-J).

Two negative results are also reported: post-hoc attribution methods disagree substantially on the same fitted model, and a global attribution ranking can remain stable while the classifier stops working (Sections V-H and V-I).

II. Related Work

Coding and speech task degradation. Reddy and Vijayarajan [1] found that standard codecs reduce SER accuracy by an amount that depends on both the codec family and the feature representation, which implies that coding interacts with the front end. In neural codec benchmarking, Wu et al. [8] showed that signal-domain fidelity metrics do not predict whether task-relevant information is retained: a codec can reconstruct a signal accurately while discarding the paralinguistic content that downstream tasks require. Signal-domain displacement and model behaviour are therefore reported here as separate quantities.

Interpretability for tabular and audio models. Post-hoc attribution has converged on local surrogates [9], Shapley-value attribution [10] with an exact algorithm for tree ensembles [11], and permutation importance; for differentiable models, path-integral [12] and reference-based [13] methods are standard, with consolidated implementations [14]. Surveys map the field [15]. Rudin [16] argues that high-stakes decisions should use intrinsically interpretable models instead of post-hoc explanations. The transfer of these methods to audio is debated. Haunschmid et al. [17] argue that audio explanations should be *listenable*, and that applying image-segmentation logic to spectrograms produces components with no perceptual correlate; related formulations appear in [18], [19]. Nasr et al. [20] extend this critique to speech emotion. Attribution over named descriptors avoids the objection by construction, at the cost of restricting the model to hand-designed features.

Attribution stability. Ghorbani et al. [2] showed that attributions can change substantially under imperceptible perturbations while predictions remain fixed. Adebayo et al. [6] showed that some saliency methods are independent of both the model parameters and the data, and Krishna et al. [7] documented frequent disagreement between explanation methods applied to the same model. Taken together, these findings indicate that attribution stability should be measured against a null rather than assumed.

Distribution shift. The mechanism identified in this study is covariate shift [21], with one unusual property: the shifted variable is not a proxy for anything the label depends on, since masking it has no cost on clean audio (Section V-J).

Research gap. Coding effects have been characterised for accuracy but not for evidence, and attribution fragility has been characterised for adversarial perturbation of images but not for routine degradation of audio. No prior work known to the author isolates the container under a verified constraint of signal invariance, or calibrates a shift in feature importance against a matched empty perturbation.

III. Corpus, Stimuli, and Features

A. Corpus

CREMA-D [22] comprises 7,442 clips from 91 actors, each delivering one of twelve fixed sentences in one of six acted emotions (anger, disgust, fear, happiness, neutral, sadness). Because the words are constant across emotions, the lexical channel is removed, and any signal the models recover must be carried by acoustic realisation, which is the property that perceptual coding alters. The CREMA-D crowd raters, hearing audio alone, matched the intended emotion on only 45.5 % of clips, with per-class recall ranging from 0.966 (neutral) to 0.164 (sad). Models trained on the intended label can exceed this figure because they exploit systematic acting cues; it is reported as context rather than as a target.

B. Reference construction and conditions

The canonical reference is 16 kHz mono 16-bit PCM, loudness-normalised to EBU R128 ($I = -23$ LUFS, LRA $=$ 7, TP $= -2$ dB). Each coded condition is produced from that reference by a single FFmpeg encoder call (`libmp3lame -b:a 64k` for MP3; `aac -b:a 64k -f mp4` for AAC), and every condition is returned to PCM through one identical decoder call, so the decoder never confounds the codec.

The fourth condition allows the codec effect to be identified. `roundtrip_wav` is the AAC bitstream of `mp4_aac64`, decoded and re-wrapped as WAV, so it carries the same codec output in a different container, with one additional int16 requantisation. Its median standardised feature difference from `mp4_aac64` is 7.6×10^{-5}, and only 12 of 436 descriptors exceed 0.05. Section V-G uses it as an informationally empty perturbation. Feature extraction over the $7{,}442 \times 4 = 29{,}768$ files failed on exactly one clip in all four conditions (`1076_MTI_SAD_XX`, documented as containing no audio), leaving 7,441 clips.

C. Feature representation

Each file is reduced to 436 named descriptors in four families. Every frame-level contour is summarised by eight order statistics (mean, standard deviation, minimum, maximum, median, interquartile range, skewness, excess kurtosis). The supplementary material defines and enumerates every column.

- **Spectral** (114): centroid, bandwidth, roll-off at 85 and 95 %, flatness, flux, entropy, slope, zero-crossing rate, RMS energy, seven-band spectral contrast, eight band-energy ratios, and high-frequency survival ratios above 4 and 6 kHz.
- **Cepstral** (240): 20 MFCCs with their Δ and $\Delta\Delta$.
- **Prosodic** (56), from Praat via Parselmouth [23]: F_0 statistics and slope, jitter (local, RAP, PPQ5, DDP), shimmer (local, dB, APQ3, APQ5, APQ11, DDA), harmonics-to-noise ratio, formants F_1–F_5 with bandwidths, intensity, voiced-segment timing, and smoothed cepstral peak prominence (CPPS).
- **Chroma** (24), plus clip duration and peak amplitude.

Spectral and cepstral descriptors are computed with librosa [24] using a 512 point FFT, a 400 sample window and a 160 sample hop (25 ms every 10 ms); MFCCs use 40 mel

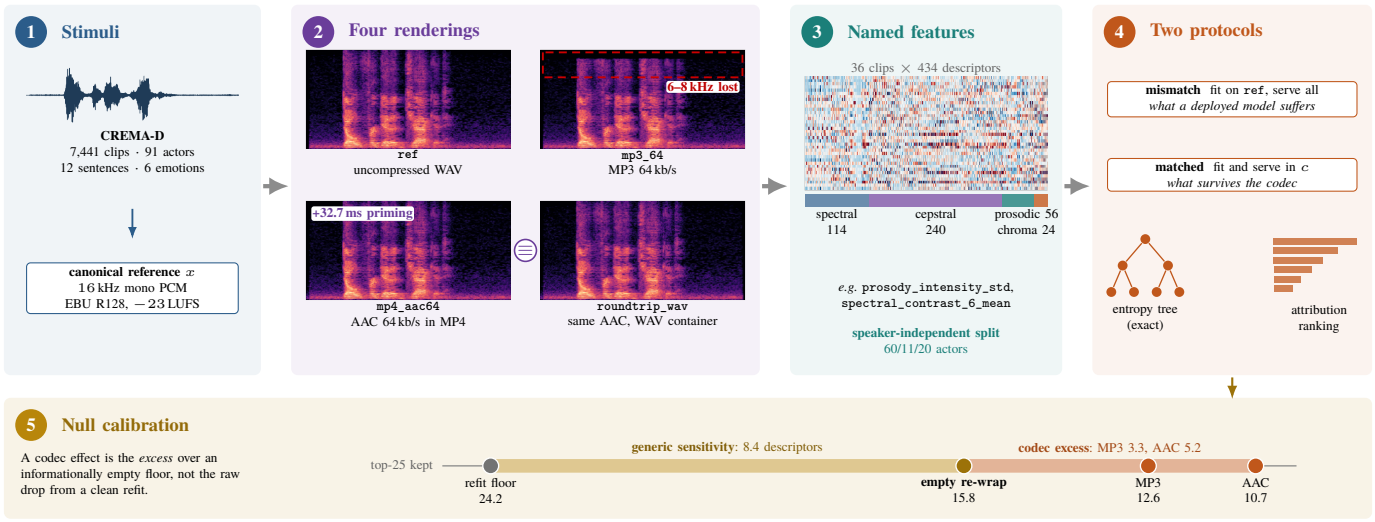


Fig. 1. Experimental design. The spectrograms show one CREMA-D clip in each of the four conditions.

bands between 20 and 8000 Hz. Praat uses a 75 Hz to 500 Hz pitch range and five formants up to 5500 Hz. Chroma tuning is pinned to zero, because an estimated tuning could itself shift under compression.

Partitioning. Actors are split 60/11/20 into training, validation and test sets with sex balance preserved (4,905, 896 and 1,640 clips), and the same assignment and the same clips are used in every condition. Speaker independence is necessary in this corpus: a variance decomposition gives a mean η^2 of 0.178 for speaker against 0.098 for emotion, and 340 of the 436 descriptors are speaker-dominant.

Separability. Of the 436 descriptors, 397 separate emotion at Bonferroni-corrected significance. The strongest are `mfcc_d1_00_std` ($F = 1194$), `mfcc_00_std` ($F = 1159$) and `prosody_intensity_std` ($F = 1078$). All three measure *variability* rather than level, a pattern that recurs in the attribution results. Fifty-one principal components are required to explain 95 % of the variance, so the feature space is high-dimensional.

D. Effect of coding on the descriptors

The displacement caused by condition c in a descriptor is measured by the standardised mean difference against the training distribution,

$$|\mathrm{SMD}_c| = |\mu_c - \mu_{\mathrm{ref}}| \,/\, \sigma_{\mathrm{train}}, \tag{1}$$

which is read in the units that matter to a model fitted on the training partition. Fig. 2 shows two distinct mechanisms.

MP3 removes most of the top octave. Perceptual bit allocation quantises the highest band coarsely and sets masked bins to zero, so `spectral_contrast_6_mean` rises from 16.38 to 46.54 and mean spectral flatness falls from 0.0189 to 0.0066. Expressed in training standard deviations, the shift is 38.6σ, which places the descriptor at inference time in a range that no model fitted on clean audio has encountered.

AAC changes frame alignment instead. Encoder priming lengthens every clip by 32.68 ms, which perturbs every frame-aligned temporal derivative slightly without removing any single band. The largest AAC shifts are therefore found in the delta-MFCC means, led by `mfcc_d1_01_mean` at SMD $= -4.16$. Delta means lie near zero and have small standard deviations, so their standardised shifts overstate the absolute change; the duration measurement provides the stronger evidence for this mechanism.

IV. Models and Evaluation

A. Classifiers

Fourteen classifiers were fitted, spanning tree ensembles [25], [26], kernel machines [27], linear and discriminant models, and instance-based and probabilistic baselines [28], together with a PyTorch MLP (widths 512–256–128, batch normalisation, ReLU, dropout 0.3) trained with AdamW (learning rate 10^{-3}, weight decay 10^{-4}, batch 128) under a cosine schedule, class-weighted cross-entropy with label smoothing 0.05, and early stopping on validation balanced accuracy (patience 25, at most 200 epochs). Missing values are imputed with the training median and inputs are standardised where the estimator requires it. For the cross-format experiments (Sections V-A–V-B) the scikit-learn models are tuned by randomised search over six candidates under three-fold cross-validation grouped by speaker and scored by balanced accuracy. The matched-format refits (Section V-D) use one fixed configuration per model in every condition, with tree models grown on the entropy criterion. Data augmentation is not used, because it would perturb the signal properties under study. The global random seed is 42.

B. Entropy and information gain

The per-condition trees of Sections V-E–V-G are grown with the Shannon-entropy criterion, so every reported split follows the rule below exactly. For a node S with class weights w_c and $w = \sum_c w_c$, the node entropy in bits is

$$H(S) \;=\; -\sum_{c=1}^{6} p_c \log_2 p_c, \qquad p_c \;=\; \frac{w_c}{w}. \tag{2}$$

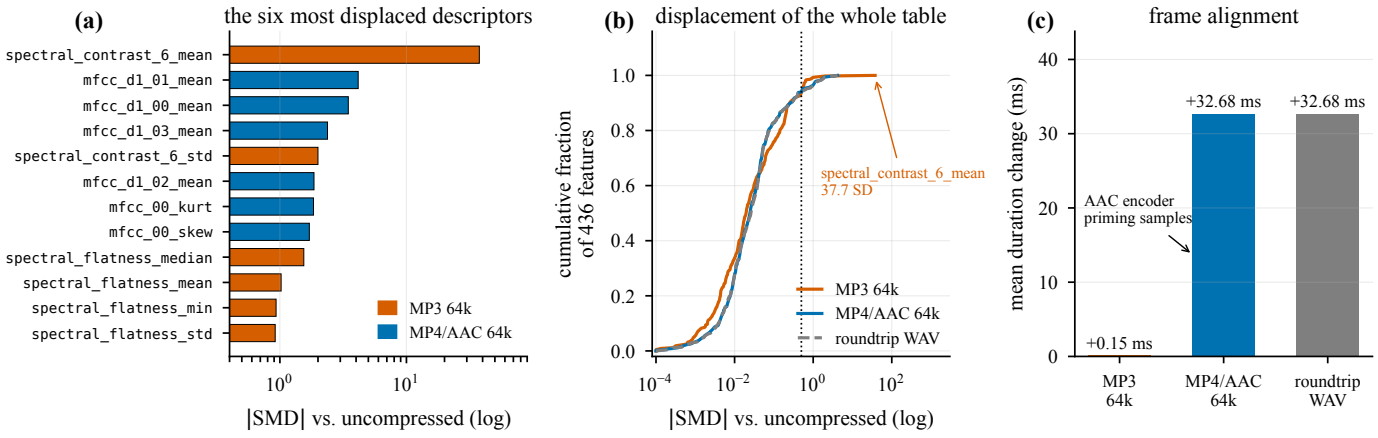


Fig. 2. Codec displacement of the 436 descriptors. (a) Most displaced descriptors per codec. (b) CDF of |SMD|. (c) Duration added by each codec.

A test on descriptor f at threshold τ partitions S into $S_L = \{x : x_f \le \tau\}$ and S_R and is scored by the information gain

$$\mathrm{IG}(S, f, \tau) \;=\; H(S) \;-\; \frac{w_L}{w} H(S_L) \;-\; \frac{w_R}{w} H(S_R), \qquad (3)$$

and the tree selects the (f, τ) that maximises it. Class weights are `balanced`, so the root entropy of every tree is $\log_2 6 = 2.5850$ bits. On uncompressed audio the best root test is `prosody_intensity_std` ≤ 8.2184, which sends 3,098.67 of the 4,905 weighted samples left with $H(S_L) = 2.3849$ and $H(S_R) = 2.0891$ bits, for IG $= 0.3090$ bits. The most informative single test on a CREMA-D clip therefore concerns the variability of its loudness, and this test resolves 12 % of the label uncertainty.

C. Agreement measures and the excess shift

An attribution E is a vector over the 436 descriptors. Two attributions are compared by *top-25 retention*, the number of descriptors shared by their 25 highest-ranked entries, and by Spearman ρ computed over the descriptors that at least one of the two fits actually uses (over all columns, ρ is inflated by the roughly 130 that both trees leave at zero). Two trees are compared by aligning internal nodes on their *path* from the root, a string of `L` and `R` branches, since the path is the only node identifier with the same meaning in two trees fitted on different data. *Descriptor agreement* at a depth is the fraction of path-aligned nodes selecting the same descriptor, and the *deepest identical level* is the greatest depth to which two trees agree at every node (0 means only the root is shared; -1 means even the root differs).

Let $A(E, E')$ be an agreement statistic, E_{ref} and E_c the attributions of models fitted on the reference and on condition c, and A_0 the agreement obtained under a manipulation that carries no coding information. The codec effect is reported as the *excess* shift

$$\Delta_c \;=\; A_0 \;-\; A(E_{\mathrm{ref}}, E_c), \qquad (4)$$

rather than as the raw drop from a clean refit, because the raw drop attributes to the codec any movement that the estimator would have produced regardless.

D. Statistical analysis

Whenever a floor and an effect are compared, both are measured within the same resampled training set. On each of 20 draws, 51 of the 60 training actors are sampled without replacement; the reference tree is fitted twice under different tie-breaking seeds ($2000 + d$ and $9000 + d$ on draw d), and each coded tree once on the same actors. Paired per-draw differences are tested with the Wilcoxon signed-rank test and reported with the paired effect size d_z. The accuracy tables are derived from a single speaker-independent split and are not tested statistically. Differences below about three points are not interpreted, and every difference on which a conclusion depends lies between 10 and 38 points.

V. Results

A. In-format performance

Table I reports performance on uncompressed audio. The spread among the top eight models is under three points, so their ordering carries little information; the differences of interest in this paper (10 to 38 points) appear only when the coding condition changes. A single decision tree, the only model that can be read in full by a person, reaches 0.391, compared with 0.599 for the network. Intrinsic interpretability therefore costs about a third of the achievable performance in this setting, which weighs against the recommendation of Rudin [16] here and in favour of post-hoc attribution on a stronger model.

B. Train/serve mismatch

Table II reports models fitted on clean audio and evaluated on coded audio. The models separate into two groups. Every tree-based model remains within 0.014 balanced accuracy of its clean score on MP3, whereas every model that uses descriptor values as continuous magnitudes loses between 12 and 38 points.

Named descriptors make the mechanism readable. A tree tests whether `spectral_contrast_6_mean` exceeds a threshold near 17, and the outcome is the same whether the value is 16.4 or 46.5, whereas a linear or kernel model propagates

TABLE I
IN-FORMAT PERFORMANCE ON UNCOMPRESSED AUDIO (20 HELD-OUT SPEAKERS)

Model	Acc.	Balanced acc.	Macro F_1	κ
ANN (PyTorch)	**0.599**	**0.599**	**0.596**	**0.518**
LightGBM	0.590	0.591	0.584	0.508
SVM (RBF)	0.585	0.586	0.581	0.502
LDA (shrinkage)	0.585	0.585	0.580	0.502
Linear SVM	0.583	0.584	0.577	0.500
XGBoost	0.581	0.582	0.575	0.497
MLP (scikit-learn)	0.573	0.574	0.569	0.488
HistGradientBoosting	0.573	0.573	0.566	0.487
Logistic regression	0.565	0.567	0.561	0.478
Random forest	0.532	0.535	0.517	0.439
AdaBoost	0.520	0.521	0.514	0.423
Extra trees	0.516	0.519	0.496	0.420
kNN ($k = 15$)	0.477	0.479	0.466	0.373
Gaussian NB	0.424	0.430	0.379	0.312
Decision tree	*0.390*	*0.391*	*0.382*	*0.268*
Dummy (majority)	0.171	0.167	0.049	0.000

the 38σ excursion directly into its decision function. The two groups are named by this property rather than by model family, because kNN and Gaussian NB apply no learned weight; in all eight magnitude-sensitive models the *magnitude* of the descriptor enters the decision continuously, while the seven tree-based models only test on which side of a threshold it falls. AAC behaves differently. It reduces the accuracy of every model, but only moderately (0.7 to 11 points regardless of family), which is consistent with the diffuse padding mechanism of Fig. 2(c). Container routing has no material effect: across the fifteen models the roundtrip and `mp4_aac64` scores differ by a median of 0.003 and at most 0.010.

C. Per-class view of the collapse

Table III breaks down the mismatched predictions of the network by class. Under MP3 the network does not predict disgust, neutral or sadness for any of the 1,640 test clips, and assigns 934 clips (57 %) to fear. Balanced accuracy records this as 0.354. Because three of the six per-class recalls are zero, the maximum attainable score for this degenerate model is 0.500, so a value of 0.354 suggests moderate degradation rather than structural failure. Macro F_1 reflects the failure more accurately (0.596 to 0.249), as does top-2 accuracy (0.805 to 0.477). AAC produces the same pattern in a milder form: neutral predictions fall from 248 to 84, and disgust absorbs most of the difference (432 against 221). The container control reproduces this pattern to within a few clips per class, which confirms that the redistribution originates in the codec and not in the decode path. A monitor of balanced accuracy would record an ordinary degradation, whereas a monitor of the predicted-class histogram, which is trivial to compute, would show half of the label space disappearing.

D. Matched-format training

The mismatch result describes deployment practice rather than the codec itself. To separate the two, each model was refitted *within* each condition on the same actors and items (Table IV). These refits use one fixed configuration per model and entropy-criterion trees, so their WAV column differs slightly from Table I. Trained on MP3, the RBF-SVM reaches 0.589 on MP3, against 0.586 on clean audio, a recovery of 38.6 points over its mismatched score of 0.203. Logistic regression recovers 33.0 points, LDA 24.8 and the shallow MLP 18.2. No model varies by more than 0.039 across the four conditions, and the two largest spreads belong to single decision trees.

Three conclusions follow. First, the codec does not remove the affective information, since a model fitted on MP3 comes within 0.003 of the clean score; the large losses in Table II therefore reflect deployment practice rather than the codec. Second, the advantage of the tree family exists only under mismatch: with matched training, LightGBM leads the RBF-SVM by 0.008 on clean audio and trails it by 0.006 on MP3. Third, the remedy is specific to one format. The RBF-SVM trained on `mp4_aac64` and evaluated on MP3 falls to 0.167, which is chance level. Matched training protects a model only against the codec on which it was trained, so the descriptor-level repair of Section V-J is more practical when the delivery format is not known in advance.

E. Per-condition tree splits

Equal accuracy does not imply equal evidence. Fig. 3 shows the top three levels of an entropy tree fitted independently on each condition, with the node entropy and information gain given at every node. The supplementary material lists every internal node to depth 4.

The primary evidence is unchanged. All four trees select `prosody_intensity_std` at the root, with an information gain of 0.3090 bits under WAV and MP3 and 0.3064 bits under AAC.

The root threshold moves in a way that matches the coding mechanism. MP3 shifts it by 0.03 % (8.2184 to 8.2210) and AAC by 1.29 % (to 8.3241). The 32.68 ms of near-silent priming increases the dispersion of the intensity contour, and the tree compensates with a higher threshold, so the padding mechanism of Fig. 2(c) is visible as a parameter of the decision rule.

The trees diverge at depth 1 under AAC but not under MP3. The WAV and MP3 trees both split the quieter branch on `mfcc_d1_00_std` (thresholds 3.9965 and 4.0175). The AAC and roundtrip trees instead select `mfcc_00_iqr`, a *static* dispersion measure of the same coefficient, with almost identical gain (0.1408 against 0.1415 bits). A frame-alignment perturbation corrupts the temporal derivatives but leaves the distribution of the contour intact, and the tree selects an equally informative test that the codec has not affected.

The container control is reproduced at the level of tree structure. `mp4_aac64` and `roundtrip_wav` select identical descriptors at all seven nodes, with a root threshold of 8.3241 in both and depth-1 thresholds of 60.5853 and 60.5802. The distinction between container and codec, previously observable only as a difference in accuracy, is thus observable in the learned decision rule itself.

F. Depth of structural agreement

Agreement at the root concerns a single node out of many. The same tree was therefore grown to depth 6 in each condition

TABLE II
BALANCED ACCURACY OF MODELS TRAINED ON UNCOMPRESSED AUDIO AND SERVED CODED AUDIO

	Model	ref	mp3_64	Δ MP3	mp4_aac64	Δ AAC	roundtrip
axis-aligned partitioning	LightGBM	0.591	0.586	−0.005	0.559	−0.032	0.562
	XGBoost	0.582	0.586	+0.004	0.560	−0.022	0.560
	HistGradientBoosting	0.573	0.559	−0.014	0.530	−0.043	0.533
	Random forest	0.535	0.532	−0.003	0.522	−0.013	0.522
	AdaBoost	0.521	0.511	−0.010	0.487	−0.034	0.484
	Extra trees	0.519	0.521	+0.002	0.512	−0.007	0.507
	Decision tree	0.391	0.382	−0.009	0.367	−0.023	0.371
magnitude-sensitive	*k*NN	0.479	0.362	−0.118	0.467	−0.012	0.457
	Gaussian NB	0.430	0.277	−0.152	0.437	+0.008	0.433
	Linear SVM	0.584	0.405	−0.179	0.471	−0.113	0.462
	MLP (scikit-learn)	0.574	0.383	−0.191	0.491	−0.083	0.494
	ANN (PyTorch)	0.599	0.354	**−0.246**	0.513	−0.087	0.512
	LDA	0.585	0.338	−0.247	0.507	−0.079	0.502
	Logistic regression	0.567	0.233	−0.334	0.472	−0.095	0.470
	SVM (RBF)	0.586	0.203	**−0.383**	0.499	−0.087	0.497

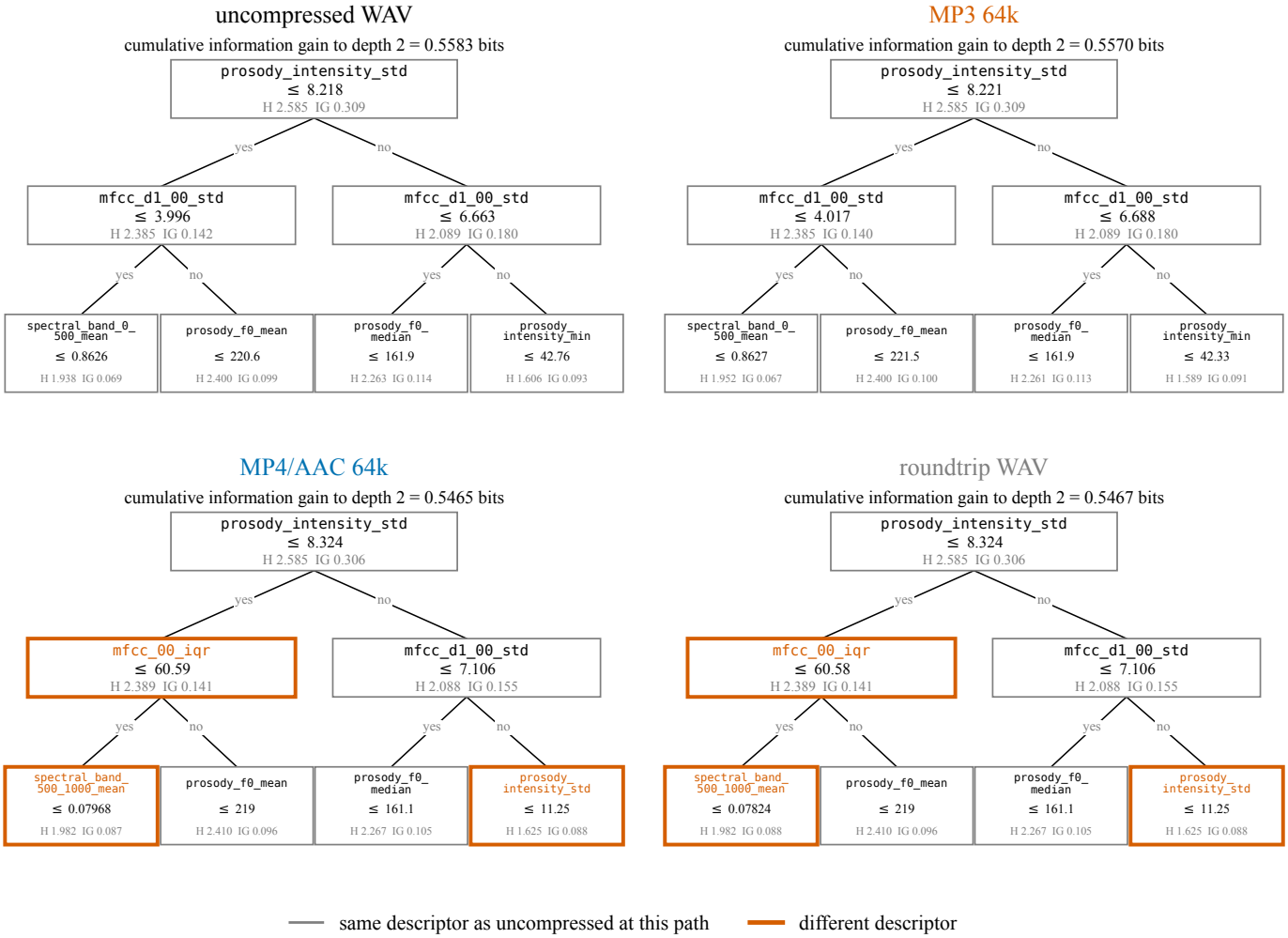


Fig. 3. Top three levels of the entropy tree fitted on each condition, with node entropy H and information gain in bits.

(55 to 61 internal nodes, with a 25-sample leaf minimum), and path-aligned nodes were compared (Section IV-C). Because tree growth is greedy, the top three levels are identical, node for node, to the trees of Fig. 3. As Fig. 4 shows, agreement decays with depth in every comparison, at a rate that depends on the codec.

The estimator itself is not the limiting factor. Refitting on identical audio with a different tie-breaking seed leaves every node to depth 4 unchanged and displaces one node of 30 at depth 5, averaged over a panel of ten seeds.

An informationally empty perturbation costs three levels of agreement. The container re-wrap reproduces the reference

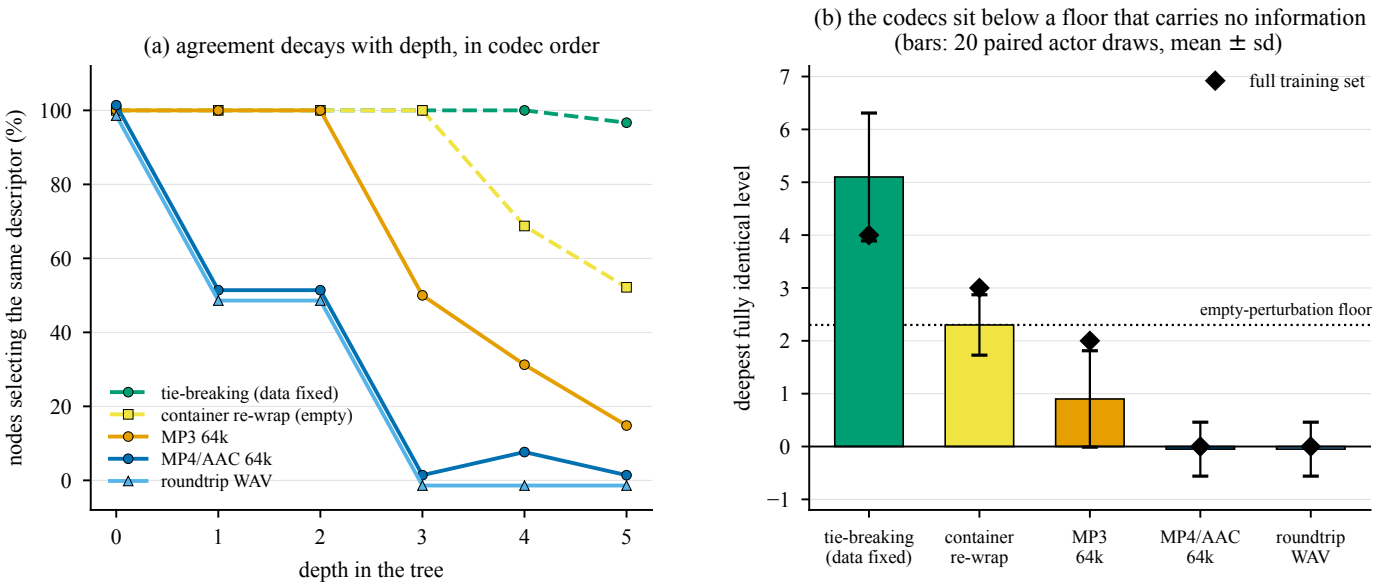


Fig. 4. Tree agreement with depth. (a) Share of path-aligned nodes selecting the same descriptor. (b) Deepest identical level over 20 paired draws.

TABLE III
NEURAL NETWORK UNDER MISMATCH BY CLASS: PER-CLASS F_1 AND NUMBER OF CLIPS PREDICTED INTO EACH CLASS

Condition	ANG	DIS	FEA	HAP	NEU	SAD
per-class F_1						
`ref`	0.770	0.467	0.555	0.567	0.611	0.607
`mp3_64`	0.660	**0.000**	0.407	0.429	**0.000**	**0.000**
`mp4_aac64`	0.718	0.416	0.555	0.538	0.340	0.481
`roundtrip_wav`	0.705	0.422	0.562	0.531	0.342	0.495
clips predicted into this class						
`ref`	299	221	322	253	248	297
`mp3_64`	375	**0**	**934**	331	**0**	**0**
`mp4_aac64`	319	432	315	292	84	198
`roundtrip_wav`	293	441	311	308	82	205

TABLE IV
MATCHED-FORMAT TRAINING: BALANCED ACCURACY WHEN A MODEL IS FITTED AND TESTED IN THE SAME CONDITION

Model	WAV	MP3	AAC	r-trip	range
LightGBM	0.591	0.583	0.582	0.585	0.008
SVM (RBF)	0.586	**0.589**	0.572	0.572	0.017
LDA (shrinkage)	0.585	0.586	0.579	0.584	0.007
MLP (scikit-learn)	0.574	0.566	0.569	0.558	0.017
Logistic regression	0.567	0.563	0.570	0.567	0.008
Random forest	0.526	0.519	0.524	0.523	0.007
Decision tree ($d = 12$)	0.408	0.379	0.398	0.369	0.039
Decision tree ($d = 3$)	0.407	0.408	0.389	0.389	0.019

tree exactly to depth 3, and then agrees on 11 of 16 nodes at depth 4 and on 12 of 23 at depth 5. Below depth 3, the structure of this tree is therefore not a property of the audio, and the calibration of Section V-G applies here as well.

Both codecs fall below that floor, in the order predicted by their mechanisms. MP3 reproduces the reference to depth 2 and AAC only to depth 0. Under paired resampling the tie-breaking floor reaches a mean deepest identical level of 5.10 ± 1.21 and the container floor 2.30 ± 0.57 ($p = 1.2 \times 10^{-4}$, $d_z = 2.06$). MP3 reaches 0.90 ± 0.91, below the container floor ($p = 3.5 \times 10^{-4}$, $d_z = 1.34$), and AAC -0.05 ± 0.51 ($p = 8.7 \times 10^{-5}$, $d_z = 2.89$). AAC displaces the structure more than MP3 ($p = 1.1 \times 10^{-3}$, $d_z = 1.15$), which reproduces the ranking result of Section V-G on a different statistic. `mp4_aac64` and `roundtrip_wav` yield the same divergence depth in all 20 draws.

The information extracted by the trees is preserved even where their structure is not. Table V evaluates Eqs. (2) and (3) at every internal node of the four depth-6 trees. Mean node entropy follows almost the same trajectory in every condition (no per-depth mean differs from WAV by more than 0.064 bits). Summed over six levels the WAV tree extracts 0.8754 of the 2.5850 available bits, MP3 0.8714 (-0.46 %) and AAC 0.8590 (-1.87 %). The container re-wrap extracts 0.8578 bits, the lowest of the four despite carrying no coding information, so differences of this size lie within the floor. Below depth 2 the AAC tree shares almost no structure with the reference, yet it resolves the same label uncertainty to within 2 %. The codec therefore changes which tests the tree selects without reducing the information those tests extract. The depth-1 substitution described in Section V-E occurs throughout the tree.

Two qualifications apply. First, the information gain per depth reaches its minimum at depth 3 and then *increases* (from 0.0832 to 0.1292 bits in the reference tree). Gains that recover at depth indicate that the tree is fitting noise in nodes of a few hundred clips, so much of the unstable deep structure should not be interpreted as evidence about emotion. Second, the thresholds of nodes that still agree diverge with depth: the median relative threshold shift under MP3 grows from 0.03 % at the root to 15.3 % at depth 5. In this feature space, codec invariance is therefore a property of the first two or three tests in the tree, which carry most of the information gain, and not of the model as a whole.

G. Null calibration of the importance ranking

Taken at face value, the importance rankings suggest a large effect: the top 25 descriptors of the tree under MP3 share only

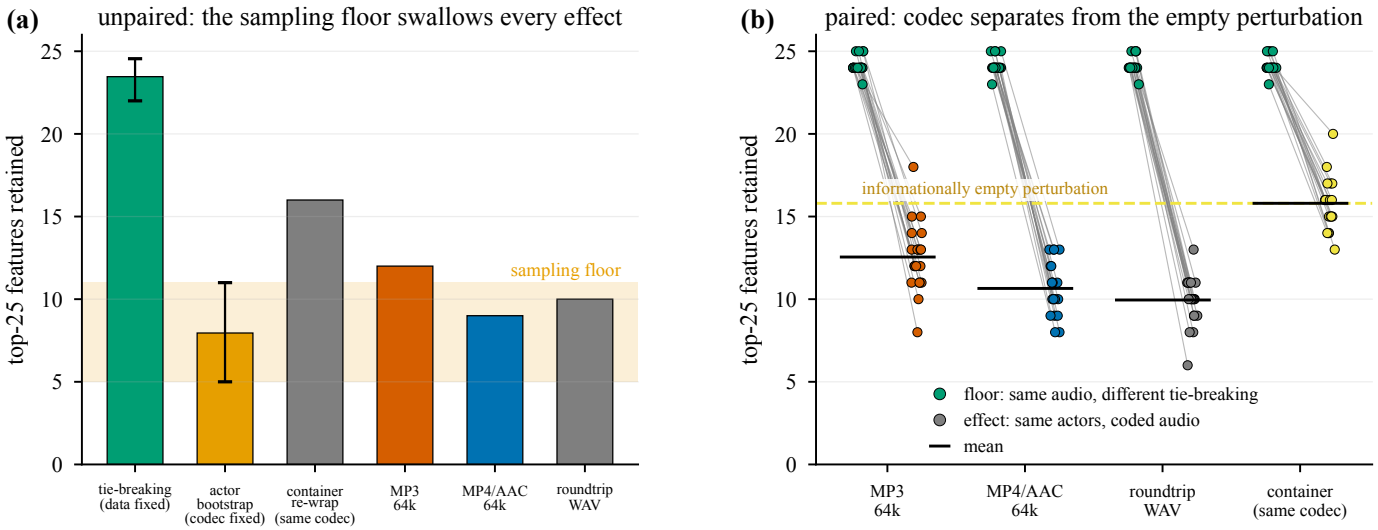


Fig. 5. Null calibration of the tree's top-25 importance ranking. (a) Marginal comparison. (b) Floors and codec effects measured within the same draw.

TABLE V
MEAN NODE ENTROPY AND SAMPLE-WEIGHTED INFORMATION GAIN PER DEPTH OF THE DEPTH-6 TREES (BITS)

Depth	ref	mp3_64	mp4_aac64	roundtrip
mean node entropy $\bar{H}$				
0	2.5850	2.5850	2.5850	2.5850
1	2.2370	2.2370	2.2383	2.2383
2	2.0516	2.0506	2.0708	2.0708
3	1.9512	1.9484	1.9427	1.9398
4	1.8547	1.8507	1.8342	1.8183
5	1.7245	1.7882	1.7458	1.7369
sample-weighted information gain				
0	0.3090	0.3090	0.3064	0.3064
1	0.1556	0.1546	0.1462	0.1462
2	0.0938	0.0935	0.0939	0.0941
3	0.0832	0.0842	0.0863	0.0860
4	0.1047	0.1027	0.1036	0.1031
5	0.1292	0.1275	0.1226	0.1220
total	**0.8754**	**0.8714**	**0.8590**	**0.8578**

TABLE VI
PAIRED WITHIN-DRAW RETENTION OF THE TREE'S TOP-25 DESCRIPTORS OVER 20 DRAWS (p FROM WILCOXON SIGNED-RANK TESTS AGAINST THE FLOOR)

	top-25 kept	ρ	p vs. floor
floor (same audio, refit)	*24.15*	$+0.958$	n/a
Container re-wrap (empty)	15.80	$+0.196$	8.1×10^{-5}
MP3 @ 64k	12.55	-0.068	8.4×10^{-5}
MP4/AAC @ 64k	10.65	-0.137	8.3×10^{-5}
Roundtrip WAV	9.95	-0.126	7.9×10^{-5}

12 members with those under WAV, with $\rho = 0.017$. This interpretation is not supported, because the deep importance ranking of a tree is fragile. Three floors were measured on the same estimator: tie-breaking (data fixed), actor bootstrap (codec fixed at WAV) and the container re-wrap (codec output fixed). In a marginal comparison, resampling the actors on which the tree is fitted retains only 8.0 of the 25 descriptors ($\rho = -0.200$), fewer than any codec condition, so every codec effect falls within the sampling floor (Fig. 5(a)). That comparison, however, varies two factors at once. Floor and effect were therefore measured within the same draw (Section IV), with the results given in Table VI and Fig. 5(b).

The estimator is reproducible: with the data fixed and only tie-breaking varied, the tree retains 24.15 of 25 descriptors ($\rho = 0.958$).

An informationally empty perturbation nevertheless has a large effect. The container re-wrap changes no codec information (its median effect on the features is 7.6×10^{-5} standardised units), yet it displaces 8.35 of the 25 top descriptors ($p = 8.1 \times 10^{-5}$). A ranking that changes by a third under a manipulation carrying no information cannot be interpreted as a description of the evidence the model requires.

The codec effect is significant but amounts to about two fifths of its apparent size. Against the empty floor of 15.80, MP3 retains 12.55 ($p = 1.1 \times 10^{-3}$, $d_z = 1.20$) and AAC 10.65 ($p = 1.2 \times 10^{-4}$, $d_z = 2.39$). Of the 13.5 descriptors AAC appears to displace relative to a clean refit, 8.35 are generic perturbation sensitivity and only 5.15 are the excess shift of Eq. (4). Reporting the uncalibrated figure would therefore overstate the effect by a factor of about 2.6. AAC displaces more of the ranking than MP3 (10.65 against 12.55, $p = 4.2 \times 10^{-3}$, $d_z = 0.84$), which reverses the accuracy ordering under mismatch: coding that slightly perturbs every temporal derivative reorganises a tree more than coding that removes a single band around which the tree can set a threshold.

H. Post-hoc attribution and its disagreements

Attribution was computed on the held-out speakers with TreeSHAP for the ensembles [11], LinearSHAP for linear models, LIME [9], permutation importance, and a depth-4 global surrogate tree. Six Captum methods [14] were applied to the network: Saliency, Input×Gradient, Integrated Gradients [12], DeepLIFT [13], GradientSHAP and Feature Ablation.

Different model families converge on similar evidence. On uncompressed audio, the top five SHAP descriptors of

TABLE VII
TOP THREE INTEGRATED-GRADIENTS DESCRIPTORS PER EMOTION, WITH ATTRIBUTION MASS

Emotion	Most attributed descriptors, and what they measure
ANG	`mfcc_d1_00_std` (0.294), `mfcc_d2_00_std` (0.232), `mfcc_00_iqr` (0.157): loudness dynamics
DIS	`duration_s` (0.285), `prosody_f2_std` (0.241), `mfcc_00_skew` (0.215): timing and F_2 movement
FEA	`prosody_f0_median` (0.235), `mfcc_d1_00_std` (0.208), `prosody_hnr_std` (0.181): raised, unstable pitch
HAP	`duration_s` (0.307), `spectral_flux_std` (0.247), `mfcc_d1_01_std` (0.175): spectral churn
NEU	`mfcc_00_skew` (0.327), `prosody_cpps` (0.327), `mfcc_d1_01_std` (0.273): steady, clean phonation
SAD	`prosody_cpps` (0.154), `mfcc_d2_00_std` (0.149), `mfcc_03_std` (0.146): breathiness

XGBoost and LightGBM are identical (`mfcc_d1_00_std`, `mfcc_d2_00_std`, `duration_s`, `prosody_intensity_std`, `mfcc_00_std`), and all six gradient methods return `mfcc_d1_00_std`, `mfcc_d2_00_std`, `prosody_cpps` and `duration_s`. The two leading descriptors measure the variability of the rate of change of overall loudness, which corresponds closely to vocal agitation. Attribution mass follows column share: MFCCs make up 55 % of the columns and receive 44 to 57 % of the attribution across models, so they have no advantage per descriptor. The decision tree relies most heavily on prosody (31 % against a column share of 13 %), which makes it readable but also limits its accuracy. The per-emotion attributions in Table VII agree with the speech literature: CPPS, the standard correlate of breathy phonation, ranks first for sadness, median F_0 for fear, and F_2 movement for disgust. Since the network received no phonetic supervision, this agreement serves as a construct-validity check on the attribution. It also indicates which emotions the codecs are likely to affect: `prosody_cpps` leaves the top 25 descriptors of the network under MP3 (Section V-I), and `duration_s`, which ranks first for disgust and happiness, is the descriptor that AAC priming shifts on every clip.

The attribution methods, however, disagree with one another. For the simpler models, SHAP agrees closely with intrinsic importance ($\rho = 0.98$ for the decision tree and logistic regression), but LIME and permutation importance are nearly unrelated to it (Fig. 6(a)). On the decision tree LIME and SHAP rank descriptors at $\rho = 0.008$ while sharing 10 of their top 20. This reproduces the disagreement problem [7] on a model simple enough to rule out approximation error as the cause. The six gradient methods, in contrast, agree almost perfectly ($\rho \geq 0.94$ for every pair; Fig. 6(b)), which reflects their common derivation rather than their correctness. Only descriptors identified by all four independent views should be reported as robust; in this study they are `mfcc_d1_00_std`, `prosody_intensity_std`, `duration_s`, `prosody_cpps` and `prosody_f0_median`.

Three further checks qualify this result. A depth-4 global surrogate reproduces the predictions of its target model in only 57 to 74 % of cases, so surrogate rules describe most of a model's behaviour but not all of it. The attributions of the network pass the model-randomisation check [6]: their correlation with attributions from a randomly initialised network of identical architecture lies between -0.011 and 0.206 for all six methods. Finally, KernelSHAP on the RBF-SVM, with 100 coalitions for 436 unknowns, is rank-deficient and diverged (maximum mean |SHAP| of 2.1×10^{12} against a median of 5.3×10^{-4}, with whole descriptor families at exactly zero). This estimate is excluded, and the SVM contributes accuracy results but no SHAP attribution.

TABLE VIII
ATTRIBUTION STABILITY UNDER MISMATCH: SPEARMAN ρ AGAINST REF, WITH TOP-25 DESCRIPTORS RETAINED IN PARENTHESES

Model	`mp3_64`	`mp4_aac64`	`roundtrip`	Δ acc. MP3
Decision tree	0.995 (24)	0.988 (21)	0.988 (21)	-0.009
XGBoost	0.991 (25)	0.984 (22)	0.984 (22)	$+0.004$
Random forest	0.990 (25)	0.976 (25)	0.975 (25)	-0.003
LightGBM	0.988 (23)	0.983 (22)	0.982 (22)	-0.005
Logistic regression	0.987 (22)	0.942 (21)	0.941 (21)	$\mathbf{-0.334}$
ANN (int. grad.)	**0.677 (11)**	0.891 (16)	0.887 (16)	-0.246

I. Attribution stability under coding

Table VIII holds each fitted model fixed and changes only its input. This differs from Section V-G, in which the model is refitted: the split structure of a fitted tree does not change when its inputs change slightly, whereas refitting can arrive at an equally good structure by a different route. Tree and linear attributions change very little, while those of the network change substantially. Accuracy nevertheless understates the change in attribution: XGBoost loses 2.2 points on AAC and swaps 3 of its top 25 descriptors, and the network loses 8.7 points and swaps 9. Under MP3, the descriptors that *enter* the top 25 of the network are almost all encoder artefacts (`spectral_flatness` in five statistics, and `spectral_contrast_6_mean`), while those that *leave* are voice-quality and prosodic measures (`prosody_cpps`, `prosody_f0_skew`, `prosody_f2_mean`, `prosody_f2_std`, `prosody_f3_bw_median`). Under mismatch, the network therefore shifts its reliance from properties of the voice to artefacts of the encoder.

The logistic regression row exposes a limitation of global attribution metrics and is the most important negative result of this study. Its SHAP ranking is essentially unchanged under MP3 ($\rho = 0.987$, 22 of 25 retained), while its balanced accuracy falls from 0.567 to 0.233. The coefficients are unchanged; only the inputs have moved out of range, and a global ranking of mean |SHAP| is largely insensitive to such movement. A global attribution-stability metric can therefore indicate no change while the classifier has effectively stopped working. Any deployment that monitors explanation stability as a proxy for model health should combine it with input-range monitoring or local attribution.

J. Diagnosis and repair

Because the descriptors are named, the failure can be corrected. Descriptors were ranked by |SMD| (Eq. (1)), and the k most shifted were replaced with their training medians at inference only, in both the coded and the clean test sets (Table IX). The single descriptor masked at $k = 1$ is

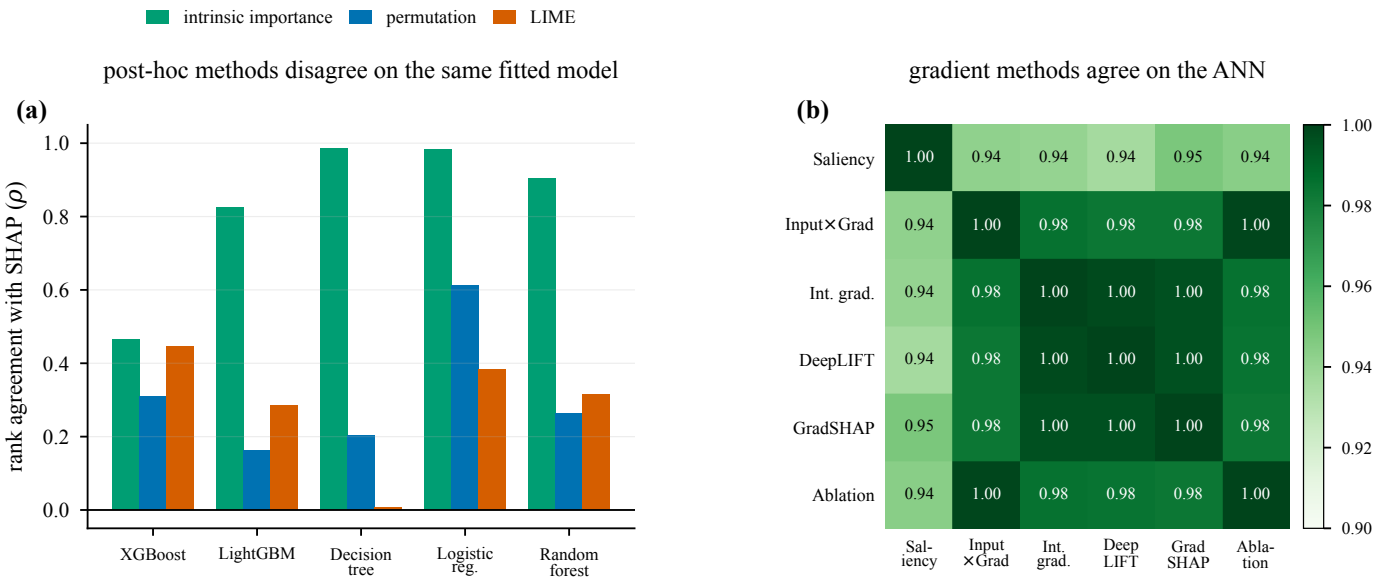


Fig. 6. Rank agreement between explanation methods. (a) Post-hoc methods on tabular models. (b) Gradient methods on the network.

TABLE IX
BALANCED ACCURACY ON MP3 AFTER REPLACING THE k MOST CODEC-SHIFTED DESCRIPTORS WITH TRAINING MEDIANS

k	SVM	Log. reg.	ANN	MLP	XGBoost	Dec. tree
clean	*0.586*	*0.567*	*0.599*	*0.574*	*0.582*	*0.391*
0	0.203	0.233	0.354	0.383	0.586	0.382
1	**0.578**	**0.502**	**0.580**	**0.552**	0.581	0.382
2	0.583	0.529	0.587	0.554	0.581	0.391
3	0.579	**0.562**	**0.595**	0.565	0.580	0.391
10	0.575	0.539	0.589	0.562	0.582	0.394
80	0.551	0.435	0.574	0.523	0.564	0.396

`spectral_contrast_6_mean`. Masking it alone recovers 98 % of the RBF-SVM's 38.3-point loss and 92 % of the network's 24.6-point loss; masking three brings logistic regression to 0.562 against a clean 0.567.

Two controls support this diagnosis. On uncompressed audio, the same masking has no cost up to 20 descriptors (the network remains between 0.607 and 0.609, slightly *above* its unmasked score of 0.599), so these columns did not carry evidence the model needed; they were the channel through which the encoder's signature entered the model. In addition, AAC behaves as its diffuse mechanism predicts: no single descriptor dominates, and the best recovery requires 10 descriptors for the network (0.513 to 0.560) and 40 for the SVM (0.499 to 0.542). The failure is therefore covariate shift on a small number of identifiable descriptors rather than a loss of emotional information, which is the conclusion reached independently in Section V-D.

VI. Discussion

A. Separating the effects of coding

The results separate four questions that a single accuracy figure combines. First, the codec does not destroy the affective information, since matched training comes within 0.003 of the clean score. Second, the codec does impair deployed models, severely so when they use descriptor magnitudes and were fitted on clean audio. Third, the codec changes what the model learns, although the leading tests are preserved: the root test is invariant, MP3 leaves two further levels intact, and the structure below is reorganised. Fourth, this reorganisation does not cost information. Below depth 2 the AAC tree shares almost no structure with the reference, yet it resolves the same label uncertainty (0.859 against 0.875 bits). The codec substitutes one test for another of equal value. This is a more specific finding than a general change in attributions, and it can be established only because the splitting criterion is explicit and the descriptors are named. Cross-format benchmarking, which addresses only the second question, would conclude that MP3 at $64\,\mathrm{kbit\,s^{-1}}$ is unsuitable for SER, when the actual problem lies in the pipeline. The per-class analysis shows the opposite risk, in which an aggregate score conceals a consequential change: a balanced accuracy of 0.354 describes a model that has become, without any visible signal, a three-way classifier.

B. The need for null calibration

The container re-wrap carries no coding information, yet it changes a third of the top-25 ranking of the tree and removes its node-for-node identity below depth 3. A claim that a manipulation changed which features a model uses cannot be interpreted without a perturbation matched in every respect except informational content. Such a control is proposed here as standard practice for feature-attribution claims in affective computing. It requires one additional condition and one additional fit, and it makes the claim falsifiable. The same control absorbed a large part of the apparent codec effect on two statistics computed from different objects (an importance ranking and a split structure), and both statistics agree on the ordering of the two codecs, which is the strongest evidence in this study that the codec-specific remainder is real. The paired design is as important as the null itself: in a marginal

comparison, the actor-resampling floor exceeds every codec effect, and no effect could have been reported.

C. Implications for deployment and auditing

Under mismatch, axis-aligned models are almost fully robust to a 38σ excursion, which is a real and inexpensive engineering advantage. That advantage disappears under matched training, however, and the learned structure of the same models is *more* sensitive to a small diffuse perturbation than to a large concentrated one. With matched training, the accuracy of the depth-6 tree differs by only 1.6 points between MP3 and AAC, whereas the depth to which each reproduces the reference structure differs by two levels. Threshold-based models therefore protect the decision but not the evidence behind it.

This distinction matters wherever per-instance rationales are presented to operators. If an audit is carried out on archived compressed material while inference runs on uncompressed input, or the reverse, the decisions will agree but the explanations will not. An explanation is reproducible only if the delivery format is recorded, and no SER benchmark known to the author reports it. A rationale based on the root test is reproducible in every condition studied here, whereas a rationale based on a rule several levels down, or on a ranked list of secondary features, is not. Rationales intended to withstand scrutiny should be limited to the depth to which the structure has been shown to reproduce for the format in use.

The repair described in Section V-J has no counterpart in a system with a learned front end, because identifying and neutralising the affected channel required units with acoustic meaning. This position is narrower than that of Rudin [16]. The results do not suggest that interpretable models should replace stronger ones, since the best model in this study is a neural network. They suggest instead that a front end based on named descriptors retains a capacity for diagnosis and repair that end-to-end pipelines lack. In this study, that capacity recovered 98 % of a 38-point loss by masking a single column.

VII. Limitations and Future Work

CREMA-D contains acted emotion on 12 fixed sentences, so the absolute accuracies do not transfer to spontaneous speech, and the human voice-only rate of 45.5 % shows that the labels are only partly recoverable from audio. The accuracy tables are based on a single speaker-independent split rather than repeated cross-validation; only the null-calibration and depth analyses use resampling. Only MP3 and AAC at $64\,\mathrm{kbit\,s^{-1}}$ were tested. The MP3 result is severe partly because this bitrate is aggressive for that codec, and Opus was not evaluated. Summary statistics discard temporal information, so no attribution reported here can localise *when* in a clip the evidence occurs. The per-condition and null analyses were designed after the cross-format results had been inspected, so the study is exploratory rather than confirmatory.

The measurement instruments have further limitations. The container control differs from `mp4_aac64` by an int16 requantisation, so it is an almost empty rather than an exactly empty perturbation. This error is conservative, because an exact null would lower the floor and enlarge the codec effects. The null calibration and the depth profile cover a single readable tree, and it remains open whether ensemble or gradient attributions require floors of the same size, although Table VIII suggests that they are more stable. The attribution comparisons are global, and Section V-I shows that such comparisons can miss a complete loss of accuracy. Finally, because of speaker dominance (η^2 of 0.178 against 0.098 for emotion), part of any importance ranking reflects which voices are present in the training set.

Several extensions follow from these limitations. A bit-exact container null, formed by two lossless containers carrying identical PCM, would tighten the floor. Extending null calibration to refitted SHAP and integrated-gradients rankings would show whether the observed fragility is specific to trees. A bitrate ladder that includes Opus would turn the two-codec comparison into a dose-response curve, for which the deepest identical level provides a convenient single-number summary. A per-instance stability analysis would address the limitation identified in Section V-I. Beyond emotion, speaker identification and voice-quality assessment depend on the spectral detail that perceptual coding discards, and the matched-format result may not transfer to those tasks.

VIII. Conclusion

This paper examined whether perceptual audio coding changes the evidence on which a speech emotion model relies, using models whose evidence consists of named acoustic descriptors. Coding does not remove the affective information: a kernel machine that reaches 0.203 on MP3 when fitted to clean audio reaches 0.589 when fitted to MP3, against 0.586 on clean audio. Cross-format losses are caused by train/serve mismatch and can be corrected either by matched training or by neutralising a single affected descriptor. Coding does change the structure a model learns. The root test and its information gain are invariant, but MP3 reproduces the reference structure only to depth 2 and AAC only to depth 0, while the total information extracted remains within 2 %.

The size of this change is easily overstated. An informationally empty container re-wrap alone displaces 8.35 of the 25 most important descriptors of a tree, and only the difference beyond that floor, 3.25 descriptors for MP3 and 5.15 for AAC, can be attributed to the codec. Claims that an intervention changed a model's explanation should therefore be accompanied by the divergence produced by an intervention that carries no information. Without that comparison such claims cannot be evaluated; with it, the codec effect reported here is reduced by a factor of about 2.6.

Data Availability

CREMA-D is publicly available from its authors [22]. The stimulus construction scripts, condition manifests, the schema of the 436-descriptor table, and the result files from which every table and figure is generated are available from the author upon reasonable request. The figure and supplementary-table generators read only these files and contain no hard-coded results. The 243 MB feature table and the rendered

audio can be regenerated from the released extraction code. All computation was performed on a CPU (Apple M5, 24 GB) with Python 3.11, scikit-learn 1.9, PyTorch 2.13, librosa 0.11, Parselmouth 0.4.7, SHAP 0.51, LIME 0.2 and Captum 0.9.

Ethics Statement

This study uses a published corpus of consenting professional actors performing scripted sentences, collected and released for research by its original authors [22]; no new recordings were made. Speech emotion recognition is a dual-use technology. The findings concern robustness and the validity of explanations, and they caution against deploying such systems on heterogeneous audio without format-aware validation. Nothing in this paper should be taken as evidence that automatic emotion inference is accurate, fair or appropriate in consequential settings.

Acknowledgments

This research received no external funding. The author used Claude (Anthropic) to assist with editing and condensing the manuscript text and with preparing the drawing code for Fig. 1. The author reviewed all such content and takes full responsibility for the article.

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